Minimum Wage Policy and Business Formation: Evidence from U.S. State-Level Data

Habib Gul Ahmadzai

ahmadzai.habib@rutgers.edu

Advanced Quantitative Methods

Final Research Paper

May 2026

Total pages: 25

Word count: ≈ 5240

Tools Utilized: Google Colab Python, Grammarly, Google Scholar, J-store, and Zotero

Abstract

The present study evaluates if there is a relationship between state-level minimum wage policy and business formation in the United States. The sample consists of 30 states from 2004 to 2024 (N = 624 state-year observations). Two-way fixed effects regressions are used with state and year fixed effects and standard errors are clustered at the state level with small sample corrections made since the number of clusters is relatively small. The measure of business formation includes total business applications from the U.S. Census Bureau Business Formation Statistics obtained from the Federal Reserve Economic Data. The preferred model incorporates real GDP, unemployment rate, personal income per capita, and population as controls. The key findings reveal that a $1 change in the state minimum wage results in a 2.7% reduction in total business applications while other variables remain unchanged ($p < .001$). The annual minimum wage change variable is positive but not significant. But after accounting for the state-specific linear trends, the estimate shrinks to about 1.5 percent, becoming marginally significant, which means that at least part of the identifying variation is likely to be correlated with state trends that change slowly. The placebo effect from a wage increase in the future yields an estimate that is only marginally significant ($p = .071$). The results are in line with a deterrence effect through the labor cost channel.

Keywords: Business Formation, Minimum Wage, Fixed Effects, Entrepreneurship, Panel Data, Robustness

Table of Contents

Abstract 0
1. Introduction 1
2. Background and Literature Review 2
  2.1 Employment Effects and the Minimum Wage Debate 2
  2.2 Firm Entry, Exit, and Profitability 3
  2.3 Demand-Stimulus Channel and Research Gap 4
3. Research Objectives and Research Questions 4
4. Data and Variables 5
  4.1 Sample and Data Sources 5
  4.2 Variable Construction and Descriptive Statistics 6
5. Econometric Model 10
  5.1 Main Specification 10
  5.2 Identification Strategy and Standard Errors 11
6. Results and Diagnostics 12
  6.1 Primary Regression Results 12
  6.2 The Identification Flip 13
  6.3 Robustness Specifications 14
  6.4 Placebo Test for Reverse Causality 15
  6.5 Model Fit and Residual Diagnostics 15
7. Interpretation of Results and Policy Implications 17
  7.1 Appropriate Causal Language 18
8. Summary, Limitations, and Future Research 18
  8.1 Summary of Findings 18
  8.2 Statistical Limitations and Honest Assessment 19
  8.3 Future Research Directions 20
References 21

1. Introduction

Minimum wage policy continues to be one of the most prominent and controversial labor market policies in the U.S. Minimum wage proponents claim that minimum wages lead to higher earnings, lower poverty rates, and increased economic security among low-wage earners. On the other hand, minimum wage opponents argue that mandated rises in labor costs would lower hiring decisions, harm company profits, or hinde entrepreneurship activities. While the empirical literature has been dominated by debates concerning employment impacts within already operating companies, employment is not the only marginal impact of minimum wage policy.

In this study, we examine the issue of firm entry, which is an equally relevant but relatively understudied topic. New firms account for most employment growth, innovation, and economic development in the long run (Meer & West, 2016). In case an increase in the wage floor leads to a reduction in new business formatiom, the impact would occur slowly through a decline in entrepreneurship rather than through lay-offs from current establishments. Through such an indirect effect mechanism, minimum wage policy could have significant macroeconomic implications in scenarios where its direct impact on employment is small.

The theoretical link between minimum wages and business start-ups is inconclusive. From a cost standpoint, minimum wages increase the anticipated payroll expense, which makes the minimum revenue requirement higher to justify profitabble business creation. The impact will be more significant on labor-intensive firms with a slim margin of profit, such as restaurants, shops, personal services, and small businesses. From the demand standpoint, the increase in wages increases the income of the affected workers, which increases the demand for goods and services. Hence, the impact of minimum wages on new business creation remains empirically ambiguous.

The sample comprises a panel of 30 states within the U.S. over the period 2004 to 2024. The 30 states indicate the number of state clusters that enhance the credibility of clustered inference and include some low-wage Southern states where the minimum wage is likely to be binding. The specification of choice is a two-way fixed effects model using state and year fixed effects, economic variables, and stateclustered standard errors with finite-sample correction. Six regressions are estimated, including lagged policy effects, state-specific linear trend effects, and future increase in minimum wage placebo test.

This paper is meant to be applied policy analysis as opposed to a prediction tool. The idea is to find out if there is any correlation between minimum wage policies and business startups after controlling for persistent differences among states and national shocks. In absence of fixed effects, the difference in structure among states overwhelms the regression process to give false coefficients like in the raw positive correlation shown in **figure 3** below.

The rest of this paper is structured in the following order. The first section is about reviewing literature. The second section deals with the objectives of ths research. The third section will discuss about data and variables. The fourth section will cover econometric model and identification. Section five will show the results and diagnostic test findings. Section six will be for interpreting findings and implications.

2. Background and Literature Review

2.1 Employment Effects and the Minimum Wage Debate

The classic economic theory suggests that if the government imposes a binding minimum wage, then employment would be reduced due to an increase in the price of labor beyond equilibrium. However, empirical studies have produced results different from those implied by the

traditional theory. Card & Krueger's study (1994), which compared employment rates at fast food outlets in New Jersey and Pennsylvania in 1992 when the state of New Jersey had just raised its minimum wage rate, revealed no evidence of any job losses.

The study refuted the conventional wisdom and showed the relevance of using quasi-experimental comparison groups in minimum wage studies. Dube et al.'s study (2010) was innovative in the sense that it compared contiguous county pairs along state boundaries, revealing the same lack of employment reduction effect on minimum wage hikes but stressing the reason for studying the effects of policies within-state variation as opposed to between states. The study by Meer & West (2016) is particularly pertinent to this paper since they focused on employment growth instead of employment levels following minimum wage hikes.

2.2 Firm Entry, Exit, and Profitability

An increasing number of studies have begun analyzing the impacts of minimum wage policy on firms rather than just employees. The exit of restaurants from the market in San Francisco was analyzed by Luca an Luca (2019). They found that increases in minimum wages increased the probability of exiting the market, especially for the low-quality firms that run close to breaking even. If marginal firms have a higher likelihood of closing their operations due to increased cost, then the new firms might not consider entering the market because the expected profitability of marginal entry would also decrease.

The introduction of the United Kingdom's National Minimum Wage was analyzed by Draca et al. (2011), who found that the affected firms became less profitable. This finding is very relevant to entrepreneurship because entrepreneurs assess whether they expect higher revenues or costs before making an entry into the market. An intervention that leads to higher expected labor costs increases the entry barrier for firms. The general impacts of minimum wage interventions

were comprehensively analyzed by Neumark and Wascher (2008), who emphasized that these impacts take place through multiple dimensions of employnent, prices, and profits.

2.3 Demand-Stimulus Channel and Research Gap

There are other plausible channels besides the cost channel. An increase in minimum wages will lead to an increase in income for workers who earn low wages. They have a higher propensity to consume compared to people with high incomes (Card & Krueger, 1994). There can be a short-term effect on the demand side from increased consumer spending. The year-over-year wage change indicator in this paper is intended to detect whether the most recent increase reflects a short-run demand or adjustment impact separate from the longer-term cost environment.

Though there have been numerous studies on the employment impacts, there is a lack of evidence in terms of business formations using a wide sample of state data in the United States. The contribution of this paper lies in its use of the two-way fixed effects model using a 30-state panel data, checking robustness through lagged policy indicators, and state-specific time trends, as well as conducting a placebo test of future wage increases.

3. Research Objectives and Research Questions

The purpose of this paper is to investigate the effect of the state minimum wage on new business creation while accounting for time-invariant characteristics of the states and common national shocks at all times. The following four research questions are proposed.

RQ1. Is there a negative relationship between the minimum wage and the number of business applications in a particular state while controlling for fixed effects and economic conditions?

RQ2. Is there an association between the yearly variation in the minimum wage and business applications that reflects any short-term demand effect that is different from the long-term cost impact?

RQ3. Is the finding robust in using High Propensiy Business Applications rather than business applications in general and using lagged specification?

RQ4. What implications can be derived from this paper for government policies, particularly those involving the tradeoff between worker well-being and entrepreneurship?

4. Data and Variables

4.1 Sample and Data Sources

The study employs a panel dataset of 30 states observed yearly from 2004 to 2024. These 30 states are California, Texas, New York, Florida, Illinois, Georgia, Kentucky, North Carolina, Pennsylvania, Ohio, Michigan, New Jersey, Virginia, Indiana, Washington, Arizona, Massachusetts, Tennessee, Missouri, Minnesota, Wisconsin, South Carolina, Colorado, Alabama, Kentucky, Louisiana, Maryland, Nevada, Oklahoma, Oregon, and Utah. The reason why the sample has been designed with 30 states is that 30 states will allow enough geographical diversity and cluster numbers. The larger sample of 30 states has been especially selected because it includes several states of the south, such as Alabama, Louisiana, South Carolina, and Tennessewhich work under the federal minimum wage floor.

The main dependent variable used in this analysis is the total number of business applications from the U.S. Census Bureau Business Formation Statistics (BFS) available on FRED (U.S. Census Bureau, 2024). These business applications are Employer Identification Numbers (EINs) applied for and not related to tax liens, estates, or administrative purposes. The monthly

numbers are added up to form an annual figure and transformed logarithmically. It is worth noting that EIN applications do not always lead to business formation, and hence the use of High-Propensity Business Applications is also utilized (U.S. Census Bureau, 2024).

The first independent variable is the minimum effective hourly wage in the United States as derived from U.S. Department of Labor data on the effective state minimum wage provided by FRED (U.S. Department of Labor, 2024). The federal minimum wage applies to those states which do not have their own minimum wage laws. Four economic control variables are utilized: real gross domestic product (U.S. Bureau of Economic Analysis, 2024); average annual unemployment rate (U.S. Bureau of Labor Statistics, 2024); personal income per capita; and population. Real GDP, personal income, population, and number of business applications are log-transformed, while the rest remain untransformed.

4.2 Variable Construction and Descriptive Statistics

Variables are created for each state-year pair. The number of monthly applications is added up to get annual figures, and the monthly rate of unemployment is averaged out to an annual figure. The variable of change in wages is the first difference of the state:

$$\Delta MinWage_{it} = MinWage_{it} - MinWage_{i,t-1} \ldots\ldots \text{(I)}$$

This variable distinguishes between the wage level, which reflects the long-term cost environment, and the year-to-year change, which might reflect short-term adjustment or demand stimulation effects. The sample used in estimation includes 624 state-year observations. The median number of years for which each state is observed is 21 years, with the minimum being 18 years.

Table 1

Descriptive Statistics

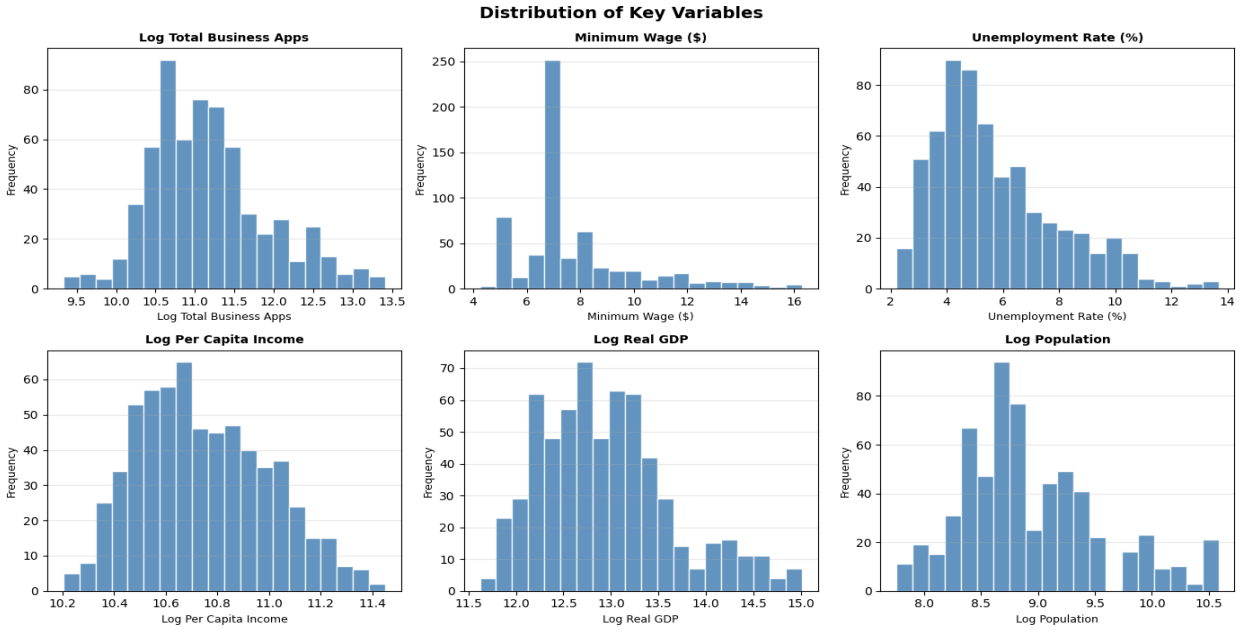
Variable	N	Mean	Std. Dev.	Min	Max
Log Business Apps	624	11.167	0.742	9.331	13.411
Minimum Wage ($)	624	7.869	2.204	4.250	16.280
Wage Change ($)	624	0.261	0.432	−0.250	2.600
Unemployment Rate (%)	624	5.725	2.201	2.217	13.675
Log Real GDP	624	12.953	0.694	11.620	15.012
Log Per Capita Income	624	10.745	0.251	10.203	11.450
Log Population	624	8.934	0.615	7.761	10.585

Note. Sample includes 30 U.S. states observed annually from 2004 to 2024 (N = 624). Log variables use natural logarithms.

The empirical distributions for the key variables are depicted in **Figure 1** below. The log transformation of the variables has greater symmetry than their levels, thereby validating the validity of the transformation method used. There is a noticeable peak around $7.25 in the minimum wage distribution due to the presence of Southern states whose minimum wage is stuck at that value.

Figure 1

Empirical Distributions of Key Variables in the 30-State Estimation Sample

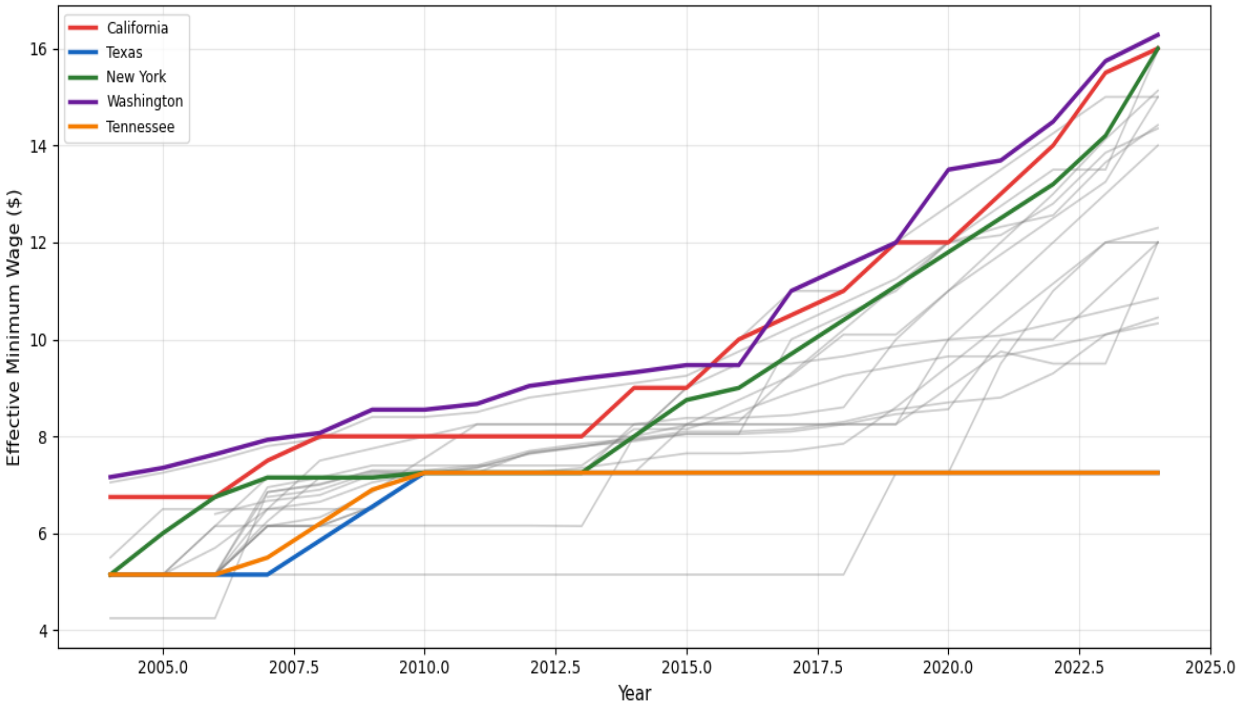


Note. Histograms of the natural log of total business applications, the minimum wage level, the unemployment rate, the natural log of per capita income, the natural log of real GDP, and the natural log of population. N = 624 state-year observations, 30 U.S. states, 2004–2024.

The path taken by the minimum wage in each of the 30 states is presented in **Figure 2** below. The states of California, Washington, New York, and Oregon experienced significant hikes in the minimum wage after 2014, whereas the states of Tennessee, Alabama, Louisiana, and South Carolina (depicted by orange and horizontal lines) stayed at the federal level through the entire period.

Figure 2

Effective Minimum Wage Trends Across 30 Sample States, 2004–2024

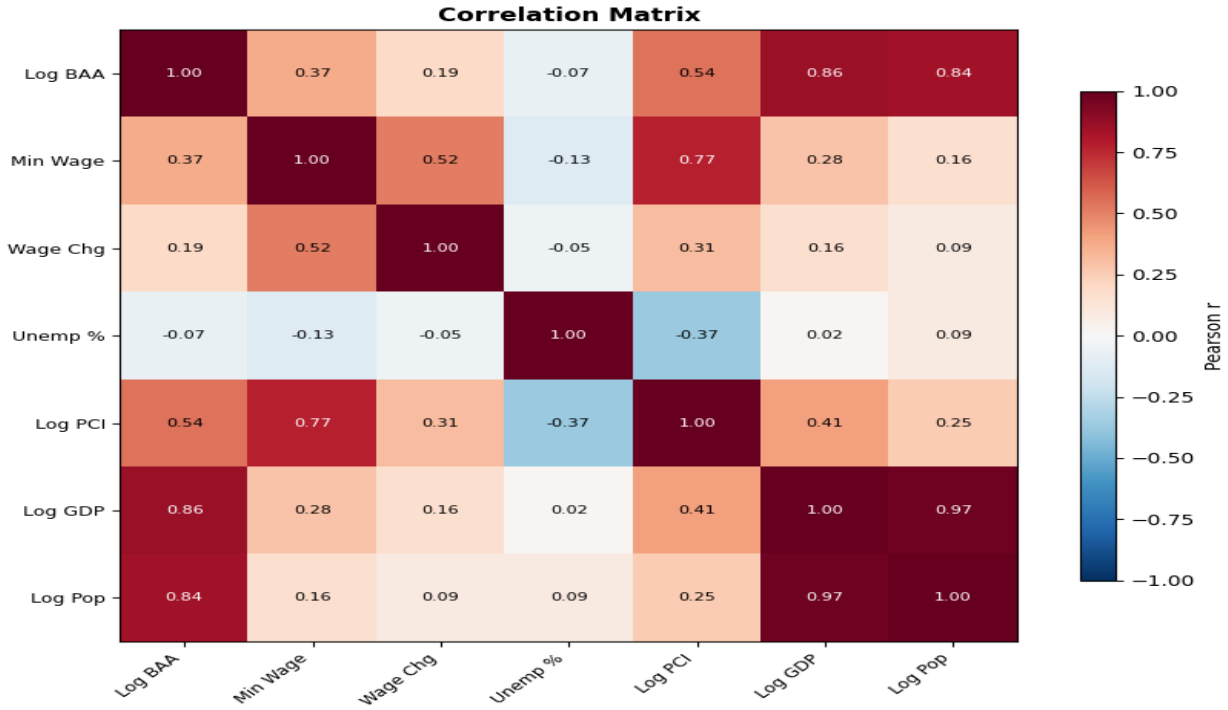


Note. Annual effective minimum wages for all 30 sample states. Selected states are highlighted: California (red), Texas (blue), New York (green), Washington (purple), and Tennessee (orange). Gray lines represent remaining states. The divergence in policy trajectories after 2014 provides within-state variation for identification in Equation 2.

Figure 3 shows the correlations of the key variables with one another using Pearson's measure. The positive value of the correlation between Log BAA and minimum wage (r = 0.37) is explained by the simple fact that wealthier, bigger states will necessarily exhibit increased business activity as well as greater wage rates, which was exactly what state fixed effects are meant to control for. The perfect correlation between Log GDP and Log Population (r = 0.97) proves the point made in Section 8.2 about multicollinearity.

Figure 3

Pearson Correlation Matrix of Key Variables



Note. Pearson correlation coefficients for all variables used in the regression models. Color scale ranges from deep red (r = +1.00) to deep blue (r = −1.00). The positive correlation between Log BAA and Min Wage (r = 0.37) motivates the fixed effects design. High correlations among Log GDP, Log Pop, and Log PCI confirm multicollinearity among the economic controls.

5. Econometric Model

5.1 Main Specification

The preferred empirical model is a two-way fixed effects regression. The full estimating equation is:

$$\ln(BusinessApps_{it}) = \beta^1 MinWage_{it} + \beta^2 \Delta MinWage_{it} + \beta^3 \ln(RealGDP_{it}) + \beta^4 UnemploymentRate_{it} + \beta^5 \ln(PerCapitaIncome_{it}) + \beta^6 \ln(Population_{it}) + \alpha_i + \lambda_t + \varepsilon_{it} \ldots\ldots (II)$$

where *i* indexes states (i = 1, …, 30) and t indexes years (t = 2004, …, 2024. α_i denotes state fixed effects that include the absorption of any time-invariant state-level factors such as geography, structural economy, and culture. On the other hand, λ_t captures year fixed effects which absorb the effects on all states of any national shock including economic recessions, the surge in entrepreneurship amid the coronavirus era, and national economic trends. β_1 is the key parameter estimate here: the percentage change in new businesses due to a one dollar increase in minimum wage.

Robustness Specification: Replace Total Applications with High-Propensity Business Applications:

$$\ln(HBA_{it}) = \beta^1 MinWage_{it} + \beta^2 \Delta MinWage_{it} + \beta^3 \ln(RealGDP_{it}) + \beta^4 UnemploymentRate_{it} + \beta^5 \ln(PerCapitaIncome_{it}) + \beta^6 \ln(Population_{it}) + \alpha_i + \lambda_t + \varepsilon_{it} \ldots\ldots \text{(III)}$$

β_1 similarity between equations 2 and 3 implies that the results do not arise from speculation and administrative use of the EIN that do not materialize into functioning firms. The lagged form entails the use of one year lag for each policy variable (equation 4), while the state trend includes a state-specific linear time trend, $\delta_i t$ (equation 5).

$$ln(BusinessApps_{it}) = \beta^1 MinWage_{i,t}{}^{-1} + \beta^2 \Delta MinWage_{i,t}{}^{-1} + \gamma' X_{it} + \alpha_i + \lambda_t + \varepsilon_{it} \ldots \text{(IV)}$$

$$ln(BusinessApps_{it}) = \beta^1 MinWage_{it} + \beta^2 \Delta MinWage_{it} + \gamma' X_{it} + \alpha_i + \lambda_t + \delta_i t + \varepsilon_{it} \ldots\ldots \text{(V)}$$

5.2 Identification Strategy and Standard Errors

This fixed-effects estimator in Equation 2 estimates β_1 based on within-state variation in minimum wages while controlling for any national shocks. This framework seeks to find out if there is a difference in the number of business applications in those years when the minimum wage

is above their historical average for that state, even after accounting for national shocks and other changes in economic factors. This approach is far more convincing compared to an approach where we compare between states with different minimum wages in cross-sectional analysis. What is required for this approach is that within-state variation in minimum wages is independent of unobserved factors that affect business applications.

Small sample cluster standard error adjustment using a small-sample t-distributed statistic for hypothesis testing will be used (Wooldridge, 2010). Using 30 clusters is toward the lower end of the 30-50 cluster guidelines suggested by Cameron & Miller (2015). The small-sample t-statistic adjustment will partly mitigate this issue. Em-dashes in the regression tables mean the variable was excluded in that specific regression.

6. Results and Diagnostics

6.1 Primary Regression Results

Table 2 presents the four primary model specifications.

Table 2

Primary Regression Results: Effect of Minimum Wage on Log Business Applications

Variable	(1) OLS	(2) FE Only	(3) FE + Controls	(4) HBA
Minimum Wage	0.1245***	−0.0282***	−0.0269***	−0.0257***
	(0.0349)	(0.0076)	(0.0062)	(0.0062)
Wage Change	−0.0015	0.0144	0.0116	0.0062
	(0.0505)	(0.0100)	(0.0093)	(0.0073)
Log Real GDP	—	—	0.3116	0.6002
			(0.3559)	(0.4096)
Unemployment Rate	—	—	−0.0185**	−0.0075
			(0.0085)	(0.0082)
Log Per Capita Income	—	—	−0.2850	0.0776
			(0.3369)	(0.3046)

Log Population	—	—	0.3118	−0.5431
			(0.5675)	(0.6847)
State Fixed Effects	No	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes	Yes
Clusters	30	30	30	30
R-Squared	0.1365	0.9916	0.9927	0.9929
N	624	624	624	624

*Note. Dependent variable is ln(Total Business Applications) in Models 1–3 and ln(High-Propensity Business Applications) in Model 4. Standard errors in parentheses; all fixed effects models use state-clustered standard errors with small-sample corrections and t-distributed test statistics. Em-dashes (—) indicate the variable was not included. *** p < .01. ** p < .05. * p < .10.*

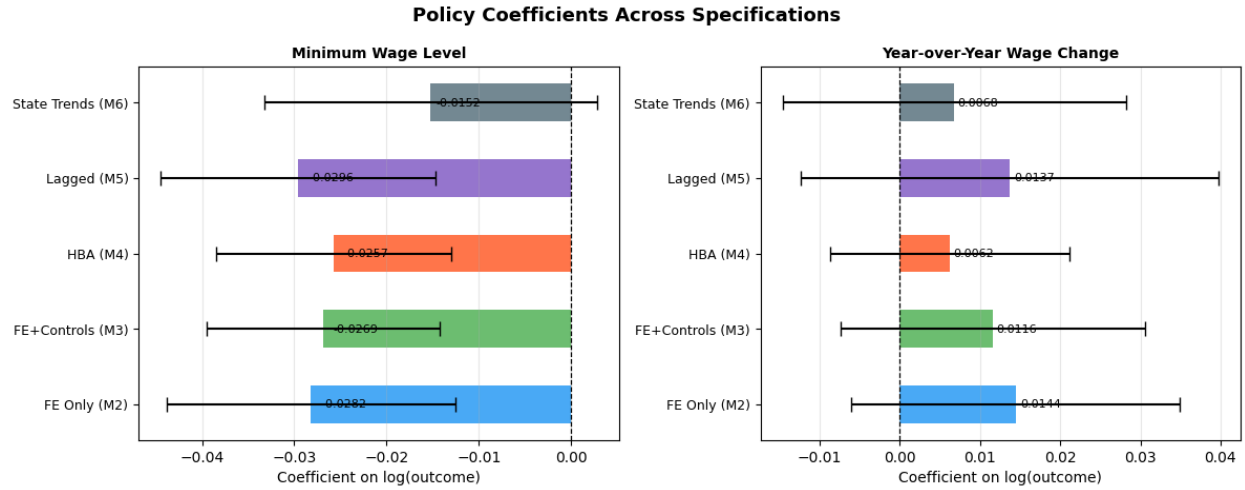
6.2 The Identification Flip

Perhaps the most critical observation is the flip in the sign of the minimum wage coefficient when moving from Model 1 to the fixed-effects models. While the coefficient on minimum wage in Model 1 is positive (+0.1245, SE = 0.0349, p < .01), as shown in **Figure 3,** the simple correlation between the natural log of business applications and the minimum wage is r = 0.37. This correlation results from the structural nature of how wealthier, larger states have both more business activity and higher minimum wages.

With state and year fixed effects included in Model 2, the estimate flips to –0.0282 (SE = 0.0076, p < .001). The sign stays the same in Model 3 (–0.0269, SE = 0.0062, p < .001) and also in Model 4, with High-Propensity Business Applications (–0.0257, SE = 0.0062, p < .001). Under Model 3, which is more likely to be correct, an increase of $1 in state minimum wages results in an estimated 2.7 percent reduction in total business applications, controlling for state fixed effects, year fixed effects, and economic variables. The estimates vary by less than 0.003 between Models 2 and 4, suggesting no overfitting. **Figure 4** shows all estimates graphically.

Figure 4

Policy Coefficient Estimates With 95% Confidence Intervals Across All Specifications



Note. Horizontal bar charts with 95% confidence intervals based on state-clustered standard errors. Left panel: minimum wage level coefficient (β_1) across five fixed effects models. All confidence intervals lie entirely to the left of zero for Models 2–5, while Model 6 (State Trends) spans zero. Right panel: year-over-year wage change coefficient (β_2). All confidence intervals span zero across every specification.

6.3 Robustness Specifications

Table 3 presents the two additional robustness specifications.

Table 3

Robustness Specifications: Lagged Policy and State-Specific Linear Trends

Variable	(5) Lagged Policy	(6) State Trends
Minimum Wage (or Lagged)	−0.0296*** (0.0073)	−0.0152* (0.0088)
Wage Change (or Lagged)	0.0137 (0.0127)	0.0068 (0.0105)
Economic Controls	Yes	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
State-Specific Trends	No	Yes
Clusters	30	30
R-Squared	0.9920	0.9957
N	594	624

*Note. Dependent variable is ln(Total Business Applications). Model 5 replaces contemporaneous minimum wage and wage change with their one-year lags; N = 594 because the lag consumes the first observation for each state. Model 6 adds a state-specific linear time trend. Standard errors in parentheses. *** p < .01. ** p < .05. * p < .10.*

Model 5 produces a lagged minimum wage estimate of −0.0296 (SE = 0.0073, p < .001), indicating that the negative relationship is robust even after accounting for timing issues. In contrast, Model 6 estimates −0.0152 (SE = 0.0088, p = .094), reduced by about 43 percent. The reduction in Model 6 is crucial because some of the identifying variation may coincide with the slow-moving path of the states. While the directio of the estimate does not varu among the six models, its magnitude is affected by state trends. As illustrated in **Figure 4** (left panel), the confidence interval in Model 6 is the only one that contains zero.

6.4 Placebo Test for Reverse Causality

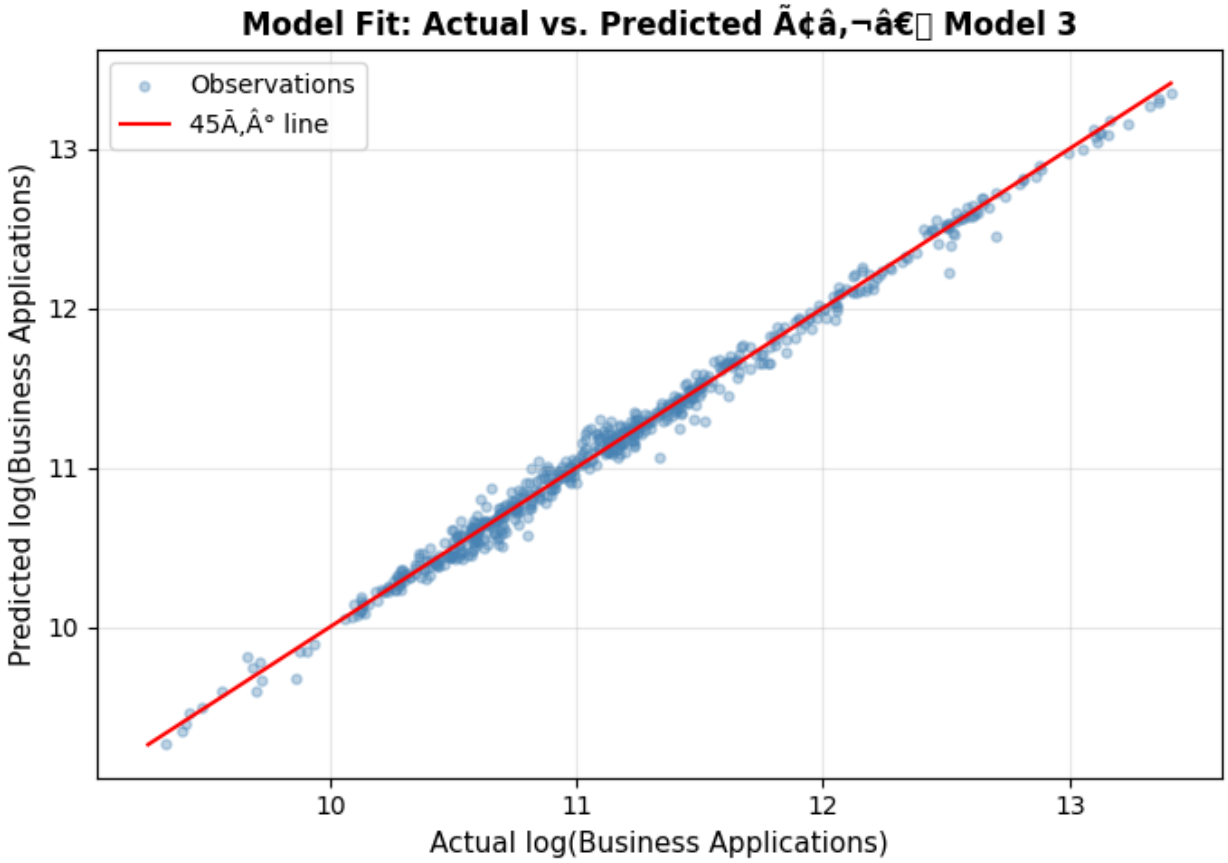
The placebo test for a future-wage increase tests whether there could be anticipatory dynamics leading to the identifying variation. The coefficient for this placebo regression is -0.0227 with p=.071. As a result, the potential exists that those states who anticipate an upcoming economic development will increase their minimum wages while at they same time lowering business creation. This placebo coefficient result is presented in full. It must be concluded that these are conditional relationships and not causal in nature.

6.5 Model Fit and Residual Diagnostics

The model with the highest goodness-of-fit statistic is one with an R-squared of 0.9927. **Figure 5** illustrates the predicted versus the actual values for all 624 data points. Data points lie very close to the 45-degree line throughout the entire empirical spectrum, indicating good fit within the sample.

Figure 5

Actual Versus Predicted Log Business Applications, Model 3

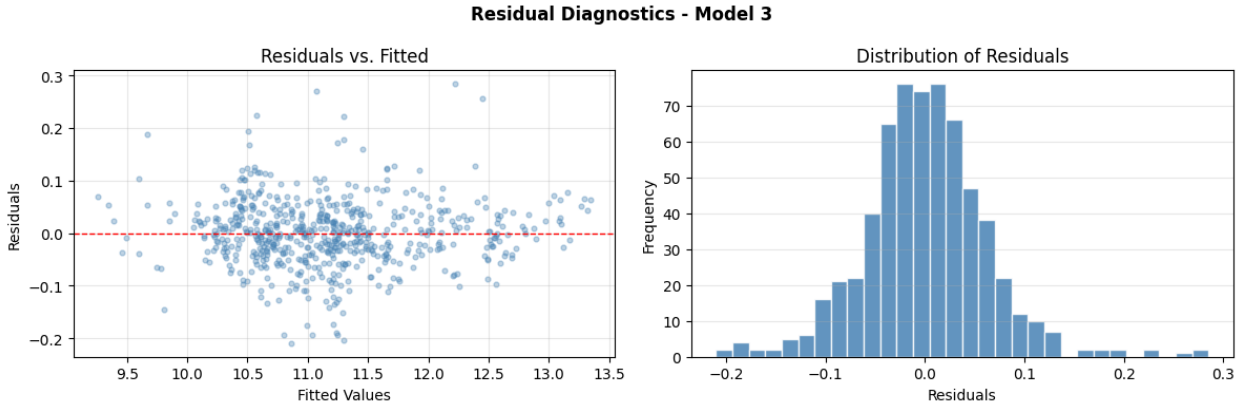


Note. Scatter plot of predicted log business applications from the preferred two-way fixed effects model (Equation 2) against observed values for all 624 state-year observations. The red line is the 45-degree reference line indicating perfect prediction.

Figure 6 shows the results of residual diagnostics. There is no pronounced systematic curve or funnel shape in the residuals, implyinhg that there is no evidence of severe nonlinearity or heteroskedasticity. The distribution of residuals is approximately normal. The average within-state AR(1) of residuals by state is 0.679, while the highest value by a state is 0.933, proving that business application data have significant autoregressive properties. This issue is accounted for using clustered standard errors, but the results warrant futur dynamic panel regressions with a lagged dependent variable.

Figure 6

Residual Diagnostics for the Preferred Two-Way Fixed Effects Model (Model 3, Equation 2)



Note. Left panel: residuals plotted against fitted values for all 624 state-year observations. The red dashed line marks zero. Residuals are centered near zero with no systematic curve or funnel pattern. Right panel: histogram of residuals. The distribution is approximately symmetric and bell-shaped. A few mild outliers are visible above +0.2 but do not dominate the distribution.

7. Interpretation of Results and Policy Implications

All four research questions are answered in the findings above. To answer the first research question, one needs only look at the persistent negative effect between the minimum wage level and business applications found in all fixed effects estimates. The preferred effect of −0.027 suggests that each additional dollar in the minimum wage leads to a decline of 2.7 percent in applications, in agreement with labor cost theory (Draca et al., 2011; Luca & Luca, 2019). On the second research question, one sees that the estimate of the wage change coefficient was insignificant regardless of specifications.

On the third research question, we can look into three different robustness checks. In terms of the HBA specification, we see an effect of −0.0257, confirming that this result is unaffected by speculation bias. On lagged regressions, the result becomes −0.0296, indicating persistence of this effect even outside the same period. Finally, for the state trend model, the estimate becomes −0.0152 and is marginally significant.

For research question four, the findings point to a clear trade-off. Minimum wage hikes may boost incomes of low-income earners. The findings show that minimum wage hikes result in fewer firm start-up applications, a factor critical in employment generation and dynamism in the economy. Policy-makers thinking about raising minimum wages need to consider other measures to offset their deterrent effects, such as streamlining business licensing, providing support to entrepreneurs at an early stage, payroll tax reduction to very small firms, startup grants in labor-intensive industries, and phased-in minimum wage hikes.

7.1 Appropriate Causal Language

It is essential to note that a good research presentation can differentiate between association and causality. In contrast to straightforward associations baseed on cross-sectional comparisons, the findings presented here are much more robust insofar as they account for fixed effects by state and year, as well as the state of the economy. Still, there is some ambiguity due to the placebo effect and weakening coefficient in Model 6. This leads us to conclude that the assumption used for identification does not hold entirely. What holds is that higher minimum wage is conditional upon lower application rates given controls.

8. Summary, Limitations, and Future Research

8.1 Summary of Findings

The analysis conducted in this paper explored the impaxt of minimum wage legislation on entrepreneurship, focusing on a panel of 30 states over the period of 2004 to 2024. In the two-way fixed-effects regression model, the point estimate for the effect of interest is −0.0269 (standard error [SE] = 0.0062, $p < .001$). In other words, there are approximately 2.7 percent fewer applications for new businesses following each one-dollar increase in the state minimum wage. The yearly wage increment is found to be positive and insignificant in all cases. The finding

remains robust for High-Propensity Business Applications and lagged independent variables. The estimated coefficient decreases to −0.0152 at marginal statistical significance for state time-trends specification. The coefficient of a wage increase to come is negative but marginally significant in the future-wages-increase placebo test (p = .071).

8.2 Statistical Limitations and Honest Assessment

Multicollinearity among economic controls is one of the important limitations in these analyses. The correlation coefficients between log GDP and log population (r = 0.97) and between log per capita income and minimum wage (r = 0.77) are seen in **Figure 3**. The former causes larger standard errors for all economic controls; the latter increases standard errors for the minimum wage variable, which is small enough (0.0062) and highly significant with a t-statistic of about 4.3. The model seems to be overfitting. However, the optimal specification includes around 59 variables for 624 observations (ratio ~1:11). The key refutation of the claim is the relative stability of the parameter of interest: β_1 varies by just 0.0013 going from Model 2 to Model 3 and 0.0027 when applying lags in Model 5. Moreover, the Small cluster count **s**houlf also be addressed**.** In this case, 30 state clusters are used, which falls at the lower bound of the 30-to-50 cluster range recommended by Cameron & Miller (2015). The small-sample adjustments made yield more stringent estimates than the asymptotic normal approximation. With a t-statistic of around 4.3, there is some headroom: an inflation of standard errors by 50 percent still yields statistical significance at the 1 percent level.

Endogeneity and the placebo warning. The future-wage-increase placebo effect is −0.0227, with p = .071. This marginal effect brings into question the possibility that states expecting a decrease in business activity will increase their minimum wages, thus inducing a

downward bias on the coefficient. Whether the sign of the bias is positive or negative is ambiguous, and the findings should be characterized as conditional relationships. Additionally, the **within-state autocorrelation**. The average within-state AR(1) fo the residuals is 0.679, and the maximum is 0.933. The business applications exhibit strong within-state autocorrelation. Including $\ln(\text{BusinessApps}_{i,t-1})$ as an additional explanatory variable is recommended in a dynamic panel setting. Finally, the State trends are a bit sensitivity. The reduction in absolute value from −0.027 in Model 3 to −0.015 in Model 6 suggests that some of the identifying variation captures slowly evolving state-level trends. In addition, the 30-state sample, which is a more realistic sample than a regional sample, does not capture all the states in the United States, especially the smallest and rural states.

8.3 Future Research Directions

There are six suggestions for improvement that can be made on the basis of the above discussion. Firstly, including all fifty states in the panel dataset would improve the statistical power as well as the number of clusters which will help address limitations 3 and 6. Secondly, adopting the contiguous county borders design by Dube et al. (2010) would enhance the identification strategy. Thirdly, employing a staggered difference-in-difference methodology would help identify pre-trends. Fourthly, introducing a lagged dependent variable would help mitigate the problem of the severe serial correlation. Fifthly, conducting the analysis at the industrial level would confirm the theoretical prediction of whether the impact is concentrated in the labor-intensive sectors.

References

Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources, 50*(2), 317–372. https://doi.org/10.3368/jhr.50.2.317

Card, D., & Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. *American Economic Review, 84*(4), 772–793.

Draca, M., Machin, S., & Van Reenen, J. (2011). Minimum wages and firm profitability. *American Economic Journal: Applied Economics, 3*(1), 129–151. https://doi.org/10.1257/app.3.1.129

Dube, A., Lester, T. W., & Reich, M. (2010). Minimum wage effects across state borders: Estimates using contiguous counties. *Review of Economics and Statistics, 92*(4), 945–964. https://doi.org/10.1162/REST_a_00039

Luca, D. L., & Luca, M. (2019). Survival of the fittest: The impact of the minimum wage on firm exit. *NBER Working Paper No. 25806*. https://doi.org/10.3386/w25806

Meer, J., & West, J. (2016). Effects of the minimum wage on employment dynamics. *Journal of Human Resources, 51*(2), 500–522. https://doi.org/10.3368/jhr.51.2.0414-6298R1

Neumark, D., & Wascher, W. (2008). *Minimum wages*. MIT Press.

U.S. Bureau of Economic Analysis. (2024). *Real gross domestic product by state* [Data set]. Federal Reserve Economic Data. https://fred.stlouisfed.org

U.S. Bureau of Labor Statistics. (2024). *State unemployment rates* [Data set]. Federal Reserve Economic Data. https://fred.stlouisfed.org

U.S. Census Bureau. (2024). *Business formation statistics* [Data set]. Federal Reserve Economic Data. https://fred.stlouisfed.org

U.S. Department of Labor. (2024). *State minimum wage laws* [Data set]. Federal Reserve Economic Data. https://fred.stlouisfed.org

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.